Membership is Ownership: A Robust Model Watermark Detection with Quantile Regression

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Abstract

Large-scale diffusion models are valuable intellectual property, yet can be stolen and adapted via downstream fine-tuning. Existing diffusion watermarking embeds explicit signatures into parameters or behavior, which may degrade utility and can be removed by post-hoc fine-tuning. We propose Membership Is Ownership, an ownership verification framework that treats a private member evidence set as a watermark and leverages the model's intrinsic memorization. Given a suspect model, we compute a multi-timestep *t-error* score by forward noising and single-step reconstruction, then aggregate with a lower quantile (Q25). To control false-positive rates, we fit a conditional Gaussian model in log-score space and obtain closed-form quantile thresholds at arbitrary target FPR. The protocol compares the suspect model against public baselines via hypothesis tests, effect sizes, and error ratios, producing statistical evidence rather than a single score. On CIFAR-10/100, STL-10, and CelebA, all datasets achieve Cohen's $d > 18$ and error ratios $> 19\times$ against public baselines; these separations persist after MMD fine-tuning (simulating post-theft adaptation), provided the adversary does not retrain from scratch.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Security and privacy** → **Digital rights management**; *Watermarking*.

Keywords

diffusion models, ownership verification, watermarking, membership inference, quantile regression

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1 Introduction

Large-scale generative diffusion models have fueled numerous profitable downstream applications. To obtain such powerful models, a huge amount of computation and data resources are consumed, which renders the trained diffusion models valuable intellectual property (IP) for tech companies and individuals. However, these IP assets are vulnerable to various unauthorized uses by adversaries seeking to leverage these models for customized, usually commercial applications.

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In practice, an adversary can obtain a checkpoint via model extraction attacks or illicit transfer, then fine-tune the model to a profitable domain or product requirement [**? ?**]. Such post-theft fine-tuning undermines the model owner's copyright and poses significant challenges for reliable ownership verification. To make a strong claim, the owner must produce evidence that is not only accurate under stringent false-positive budgets, and also valid after realistic fine-tune or adaptation.

Most existing diffusion IP protection methods embed explicit signatures into model parameters or output behavior [**? ? ? ?**]. While effective in controlled settings, such watermarks may impose utility costs and can be weakened by fine-tuning because the model thief can use every means to remove such explicit watermarks. Recently, an alternative watermark method has emerged, which is to leverage the model's intrinsic memorization [**? ? ?**] , where the reconstruction error of diffusion models on training examples is systematically lower than on non-training samples. However, as the models' generalization capability increases, the difference between training samples and non-training samples becomes marginal, resulting in the "low error-based" method being insufficient for an ownership claim. Moreover, most modern diffusion models may also use substantially overlapped training data from the public Internet, which increases the difficulty of accurate attribution.

These practical challenges necessitate a robust watermark detection method that can distinguish the IP-protected model from public reference models under a statistically controlled decision rule. To fill this research gap, in this work, we propose **Membership is Ownership**, an ownership verification framework that can accurately detect and verify whether a suspect model in the thief's possession is illegally obtained and fine-tuned from the model owner's base model, using a designated owner signature. At the heart of our approach lies a simple but principled observation: a model retains systematic memory of its training data, manifesting as lower reconstruction error on seen examples. We exploit this signal by constructing a private evidence set, cryptographically signed by the model owner, which serves as a persistent watermark without modifying the model itself. To extract a robust signal from a suspect model, we introduce a multi-timestep *t-error* score—computed via forward noising followed by single-step reconstruction—and summarize it through a lower quantile (`Q25`) to suppress noise while preserving the memorization signature. Rather than reporting a single black-box decision, we anchor the verification protocol on a principled statistical foundation: a conditional Gaussian model fitted in log-score

space yields closed-form quantile thresholds at arbitrary false-positive rates, enabling transparent and auditable comparisons against public baselines through hypothesis tests, effect sizes, and error ratios. The resulting separations are striking. Against public baselines, we observe Cohen's $d > 18$ and error ratios exceeding 19×, margins that remain robust even under MMD fine-tuning, which simulates realistic post-theft adaptation, provided the adversary stops short of full retraining or targeted memorization erasure. The paper unfolds the following contributions.

- We introduce a training-free ownership verification framework for diffusion models that leaves model parameters and output quality entirely intact, relying solely on a private evidence set as a non-invasive watermark.
- We propose a multi-timestep t-error score aggregated via lower quantile (`Q25`), which suppresses per-example noise while preserving the memorization signal, yielding a stable and discriminative per-sample statistic.
- We derive closed-form quantile thresholds under a conditional Gaussian model in log-score space, enabling precise false-positive rate control at any target significance level without requiring empirical calibration.
- We demonstrate Cohen's $d > 18$ and error ratios exceeding 19× across four datasets, with separations remaining robust under MMD fine-tuning, establishing strong empirical guarantees against realistic post-theft adaptation.

2 Related Work

Model Watermarking. Prior machine learning model watermarking methods typically embed an identifiable pattern into model behavior, most notably through trigger-based backdoors or carefully constructed inputs that induce a predetermined response [**? ? ? ?**]. Such approaches provide a direct verification interface when the watermark remains intact, but they generally require explicit intervention during training and can be sensitive to subsequent model editing or "cleaning." *In contrast to embedding an external trigger, our setting treats a private member evidence set as the "watermark" and derives evidence from the model's intrinsic memorization behavior, avoiding modifications that introduce new behavioral artifacts.*

Membership Inference Attacks. Membership inference attacks (MIA) aim to decide whether an input was part of a model's training data [**? ? ?**]. For diffusion models, recent work instantiates MIA using reconstruction-based signals [**? ? ?**], loss/likelihood-based criteria [**? ?**], or scalable quantile-regression formulations [**? ?**]. These methods are primarily evaluated as per-example decision rules (binary classification or scoring) and are typically reported via attack metrics. Ownership verification, however, requires comparative, population-level evidence: the ownership claim must be supported by aggregate separation on a designated evidence set, calibrated against reference models that represent the null hypothesis, and summarized via statistically interpretable tests rather than single-sample predictions. *Our framework uses MIA-style signals as the underlying measurements, but creates an auditable verification procedure with public references and formal decision criteria.*

3 Problem Setup

3.1 Background and Problem Formulation

Diffusion Models. A diffusion model defines a forward noising process and a learned reverse process. Given a clean sample $x_0 \sim q(x_0)$, the forward process produces a noisy version at timestep $t \in \{1, \dots, T\}$:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise schedule. A neural network ϵ_θ is trained to recover the added noise by minimizing:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,\, \epsilon,\, t}\big[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\big]. \tag{2}$$

At inference, new samples are drawn by iteratively denoising from $x_T \sim \mathcal{N}(0, I)$ using the learned ϵ_θ.

It has been empirically established that diffusion models memorize portions of their training data [**? ?**]: the reconstruction error on training examples is systematically lower than on unseen samples drawn from the same distribution. This memorization gap persists across architectures and dataset scales, and serves as the foundational signal that our framework exploits for ownership verification.

From Membership Inference to Ownership Verification. Membership inference attacks (MIAs) leverage the memorization gap to determine whether a single query sample belongs to the training set of a target model [**? ?**]. We state this formally:

Definition 3.1 (Membership Inference Attack). Given a target model M and a query sample x, a membership inference attack is a binary classifier $\mathcal{A} : (M, x) \to \{0, 1\}$ that predicts whether $x \in \mathcal{D}_{\text{train}}$.

However, low reconstruction error on a single sample does not constitute evidence of ownership, as the sample may simply be inherently easy to reconstruct for any model. Ownership verification demands a stronger, population-level claim: the suspect model must exhibit statistically significant memorization on a *designated evidence set* when compared against public baselines that share no training overlap with that set.

Definition 3.2 (Ownership Verification). Let $\mathcal{M}_A$ denote a base model trained by the owner, $\mathcal{M}_B$ a suspect model held by the adversary, and $\mathcal{W} = \{x_1, \dots, x_{|\mathcal{W}|}\}$ a private evidence set signed by the owner such that $\mathcal{W} \subset \mathcal{D}_{\text{train}}(\mathcal{M}_A)$. An ownership verification protocol is a hypothesis test $\mathcal{V} : (\mathcal{M}_B, \mathcal{W}, \{M_{\text{ref}}^{(i)}\}) \to \{\texttt{owned}, \texttt{independent}\}$ that determines whether $\mathcal{M}_B$ is derived from $\mathcal{M}_A$ by comparing the reconstruction behavior of $\mathcal{M}_B$ on $\mathcal{W}$ against a set of public baseline models $\{M_{\text{ref}}^{(i)}\}_{|\mathcal{T}|=1}^{|\mathcal{T}|}$ at a controlled false-positive rate α.

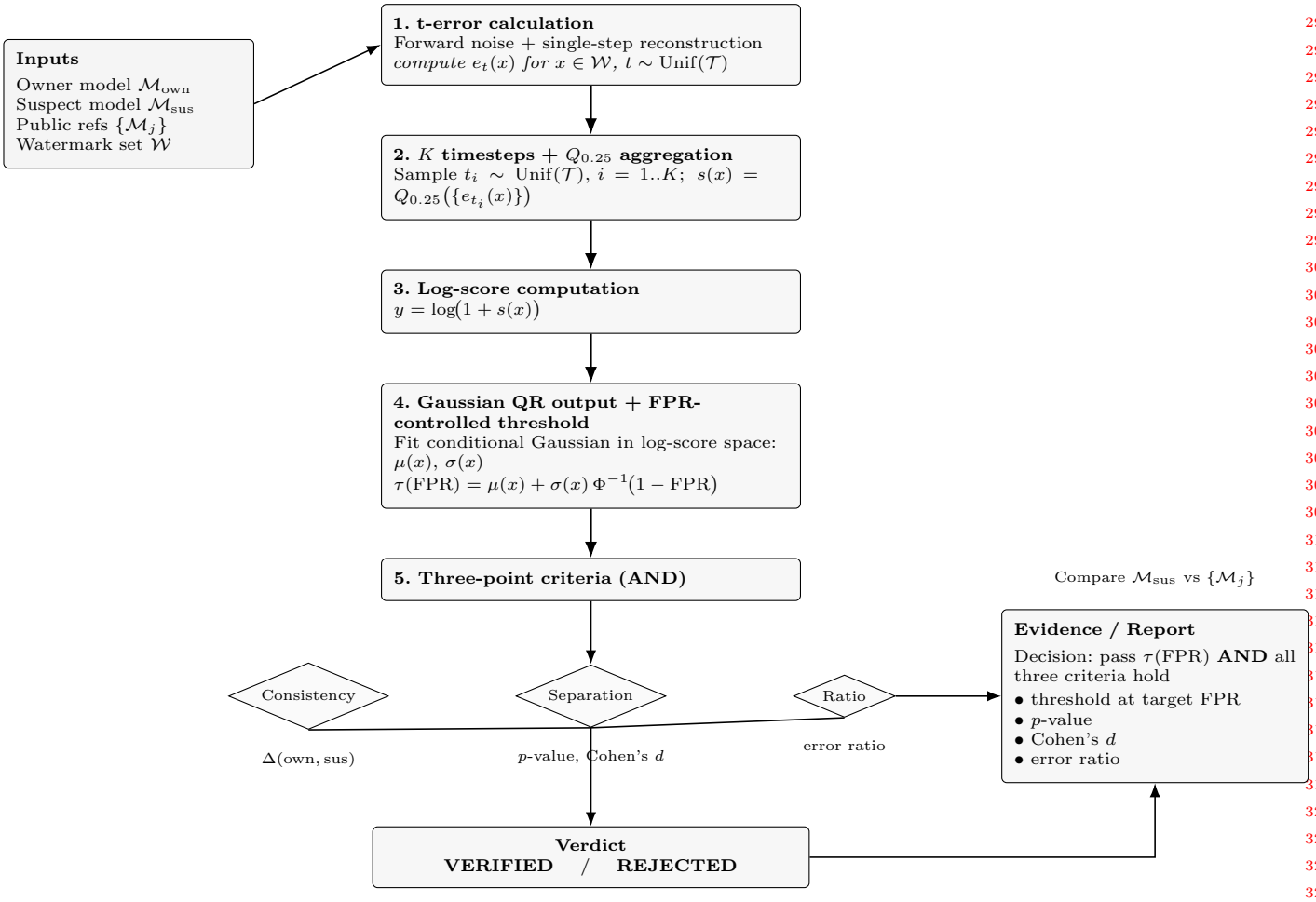

Figure 1: System overview. A private evidence set $\mathcal{W}$ serves as a watermark. We compute multi-timestep t-error scores, aggregate with a robust lower quantile to obtain $s(x)$, and fit a conditional Gaussian model in log-score space to derive a closed-form threshold at a target FPR. Ownership is supported via population-level evidence against public reference models using consistency, separation, and ratio criteria.

This formulation shifts the task from sample-level binary classification to a calibrated, population-level statistical test, which is the protocol we develop in the following sections.

3.2 Threat Model

We consider three principals in the ownership verification setting. The **owner** O trains a model $\mathcal{M}_A$ on $\mathcal{D}_{\text{train}}$, which contains a private watermark set $\mathcal{W} \subset \mathcal{D}_{\text{train}}$ known only to O. The **adversary** obtains $\mathcal{M}_A$ through theft or leakage, and produces a derivative model $\mathcal{M}_B$ by fine-tuning; $\mathcal{M}_B$ is the suspect model submitted for verification. The **verifier** V, given $\mathcal{M}_A$, $\mathcal{M}_B$, a set of public baseline models, and the private set $\mathcal{W}$, determines whether the ownership claim is valid.

We assume the adversary has white-box access to $\mathcal{M}_A$ and may possess in-domain data disjoint from $\mathcal{W}$. The adversary is aware that a verification protocol exists but cannot identify $\mathcal{W}$: beyond the owner's secrecy, even if $\mathcal{D}_{\text{train}}$ were leaked, selecting the exact $|\mathcal{W}|$-subset from N training samples is feasible with probability at most

$$P(\text{exact match}) = \frac{1}{\binom{N}{|\mathcal{W}|}} \leq \left(\frac{|\mathcal{W}|}{N}\right)^{|\mathcal{W}|}, \tag{3}$$

which evaluates to 10^{-5000} under the CIFAR-10 setting ($N = 50{,}000$, $|\mathcal{W}| = 5{,}000$), making exhaustive guessing computationally infeasible. This bound rests on a counting argument rather than a cryptographic assumption and does not claim resistance to side-channel leakage of $\mathcal{W}$ itself. The adversary is permitted to apply fine-tuning, distillation, and pruning to $\mathcal{M}_A$; full retraining from scratch falls outside our scope, as it severs provenance entirely.

Upon receiving a dispute, the verifier proceeds as follows: it first checks that SHA-256(sort($\mathcal{W}$)) matches a hash commitment timestamped prior to the dispute; it then computes

t-error scores on $\mathcal{W}$ for both the suspect model and public baselines; and finally evaluates the three-point criteria (Section 4.4), outputting VERIFIED or REJECTED together with the supporting statistical evidence (p-values, Cohen's d, and error ratio). The verifier is assumed to have white-box access to both models, consistent with standard practice in IP litigation, regulatory audits, and contractual audit clauses.

4 Method

The verification framework has four components: (1) a *t-error* score measuring reconstruction quality via forward-reverse diffusion, (2) multi-timestep aggregation for stable per-sample scores, (3) Gaussian quantile regression for FPR-controlled thresholds, and (4) a three-point statistical protocol comparing the suspect model against public baselines.

4.1 t-error Scoring

Diffusion models reconstruct their training data with lower error than unseen data [?]. We exploit this *memorization gap* by measuring reconstruction quality at a given noise level t.

Specifically, given a clean sample x_0, we first corrupt it via the forward diffusion schedule:

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

where x_t is the noised image at timestep $t \in \{1, \ldots, T\}$, ϵ is i.i.d. Gaussian noise, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative noise coefficient under a cosine schedule [?]. The model then predicts the added noise $\hat{\epsilon}_\theta(x_t, t)$, from which we recover a single-step reconstruction:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{5}$$

We define the *t-error*, denoted $e_t(x_0)$, as the per-pixel mean squared error between the original and reconstructed images:

$$e_t(x_0) = \frac{\|x_0 - \hat{x}_0\|_2^2}{H \times W \times C} \tag{6}$$

where H, W, C denote the image height, width, and number of channels, respectively. The key property underlying our framework is the *memorization effect*: models trained on a particular dataset exhibit systematically lower reconstruction error on their training examples, i.e., $\mathbb{E}_{x \in \mathcal{D}_{\text{train}}}[e_t(x)] < \mathbb{E}_{x \notin \mathcal{D}_{\text{train}}}[e_t(x)]$. This persistent gap, which survives across architectures and dataset scales, provides the foundational signal for our ownership verification protocol.

4.2 Multi-Timestep Aggregation

A single timestep provides a noisy estimate of reconstruction quality, as the magnitude of $e_t(x)$ varies substantially with the noise level t. To obtain a stable per-sample statistic, we aggregate t-errors over K uniformly sampled timesteps into a single score $s(x)$ via the 25th percentile:

$$s(x) = Q_{0.25}\big(\{e_{t_i}(x)\}_{i=1}^{K}\big) \tag{7}$$

We use K=50 by default. The choice of a lower percentile is deliberate: it captures the timesteps where the memorization gap is most pronounced while remaining robust to outlier timesteps that introduce high but uninformative error. Ablations in Section 5.5 confirm that the 25th percentile yields the strongest separation among common aggregation strategies, and that performance is stable for $K \geq 25$. The full scoring procedure is summarized in Algorithm 1.

Algorithm 1 t-error Computation

Require: Model $\mathcal{M}$, sample x_0, timesteps $\mathcal{T} = \{t_1, \ldots, t_K\}$, noise schedule $\{\bar{\alpha}_t\}$
Ensure: Aggregated score $s(x_0)$
  `errors` $\leftarrow []$
  for $t \in \mathcal{T}$ **do**
    $\epsilon \sim \mathcal{N}(0, I)$
    $x_t \leftarrow \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$
    $\hat{\epsilon} \leftarrow M_\theta(x_t, t)$
    $\hat{x}_0 \leftarrow (x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{\epsilon}) / \sqrt{\bar{\alpha}_t}$
    $e_t \leftarrow \|x_0 - \hat{x}_0\|_2^2 / (H \times W \times C)$
    `errors`.append(e_t)
  end for
  return $s(x_0) \leftarrow Q_{0.25}(\texttt{errors})$

4.3 Gaussian Quantile Regression

Rather than relying on fixed thresholds, we model the *conditional score distribution* for non-members and derive decision boundaries in closed form. Concretely, we apply a log transformation to the aggregated score and parameterize the result as a Gaussian with input-dependent mean $\mu(x)$ and standard deviation $\sigma(x)$:

$$y = \log(1 + s(x)) \sim \mathcal{N}\big(\mu(x),\, \sigma^2(x)\big) \tag{8}$$

where $\mu(x)$ and $\sigma(x)$ are jointly predicted by a neural network f_ψ. The network uses a ResNet-18 backbone to extract image features, which are concatenated with a compact statistical summary $\phi(x) \in \mathbb{R}^3$ consisting of the mean, standard deviation, and ℓ_2 norm of the t-error sequence. An MLP head maps the combined representation to $(\mu, \log \sigma)$.

We train f_ψ by minimizing the Gaussian negative log-likelihood over N auxiliary non-member samples:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(y_i - \mu_i)^2}{2\sigma_i^2} + \log \sigma_i \right] \tag{9}$$

where $y_i = \log(1 + s(x_i))$ is the observed log-score and (μ_i, σ_i) are the corresponding network predictions. Because the conditional distribution is Gaussian, the $(1-\alpha)$-quantile at any target false-positive rate α admits a closed-form expression:

$$\hat{q}_\tau(x) = \mu(x) + \sigma(x) \cdot \Phi^{-1}(\tau), \quad \tau = 1 - \alpha \tag{10}$$

with Φ^{-1} denoting the standard normal inverse CDF. This is a key practical advantage over pinball-loss quantile regression, which requires training a separate model for every desired quantile level; here, a single model supports arbitrary τ without retraining.

To translate the predicted quantile into a membership decision, we define the *margin* $m_\tau(x)$ as the gap between the

predicted quantile and the observed log-score:

$$m_\tau(x) = \hat{q}_\tau(x) - \log(1 + s(x)) \tag{11}$$

A positive margin indicates that the sample's reconstruction error falls below what would be expected of a non-member, suggesting likely membership in the training set.

Finally, to reduce prediction variance, we wrap the above procedure in a bagging ensemble of B=50 models, each trained on a bootstrap sample drawn with replacement from 80% of the auxiliary data. At inference, we average the individual quantile predictions into a single ensemble estimate:

$$\hat{q}_\tau^{\text{ens}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{q}_\tau^{(b)}(x) \tag{12}$$

where $\hat{q}_\tau^{(b)}(x)$ is the quantile from the b-th ensemble member. The ensemble margin $m_\tau^{\text{ens}}(x) = \hat{q}_\tau^{\text{ens}}(x) - \log(1+s(x))$ serves as the final per-sample statistic; as before, positive values indicate likely membership. Ensembling is embarrassingly parallel and introduces negligible additional inference cost (Section 5.6).

4.4 Three-Point Verification Criteria

Given the owner model $\mathcal{M}_A$, the suspect model $\mathcal{M}_B$, a set of public baselines, and the watermark set $\mathcal{W}$, we require all three of the following criteria to be satisfied before declaring ownership verified.

Criterion 1: Consistency. If $\mathcal{M}_B$ is indeed derived from $\mathcal{M}_A$, the two models should exhibit similar reconstruction behavior on $\mathcal{W}$. We operationalize this by requiring that a two-sample t-test on their score distributions yields $p > 0.05$, i.e., we cannot reject the null hypothesis that the two models share the same t-error distribution. Failure here suggests that the suspect model has no provenance relationship with the owner model.

Criterion 2: Separation. The owner model must reconstruct watermark samples significantly better than any public baseline. We require both statistical significance ($p < 10^{-6}$, to control for multiple baseline comparisons) and practical significance (Cohen's $d > 2.0$, corresponding to a "very large" effect size [**?**]). This dual requirement guards against claims that achieve a low p-value purely through large sample size without meaningful effect.

Criterion 3: Ratio. As a final, interpretable sanity check, we require the ratio of mean baseline score to mean owner score to exceed a conservative threshold:

$$\frac{\bar{s}_{\text{baseline}}}{\bar{s}_{\text{owner}}} > 5.0 \tag{13}$$

where $\bar{s}$ denotes the sample mean of aggregated t-error scores. This threshold is deliberately conservative; in all experiments the observed ratio exceeds 19× (Table 2).

Ownership is declared **verified** only when all three criteria are jointly satisfied. This conjunction ensures that no single metric can produce a false positive: consistency confirms provenance, separation establishes that the memorization signal is genuine rather than coincidental, and the ratio provides a human-interpretable magnitude check. The complete verification protocol, including hash commitment validation, is formalized in Algorithm 2.

Algorithm 2 Ownership Verification Protocol

Require: Models M_A, M_B, baselines $\{M_{\text{ref}}^{(i)}\}$, watermark set $\mathcal{W}$, hash commitment H_{claimed}
Ensure: VERIFIED or REJECTED
  $H_{\text{computed}} \leftarrow \text{SHA-256}(\text{sort}(\mathcal{W}))$
  if $H_{\text{computed}} \neq H_{\text{claimed}}$ **then**
    return REJECTED
  end if
  $S_A \leftarrow \{s_{M_A}(x) : x \in \mathcal{W}\}$; $S_B \leftarrow \{s_{M_B}(x) : x \in \mathcal{W}\}$
  $S_{\text{ref}} \leftarrow \{s_{M_{\text{ref}}}(x) : x \in \mathcal{W}\}$ for best baseline
  O1: $p_1 \leftarrow \text{TTest}(S_A, S_B)$; **if** $p_1 < 0.05$ **return** REJECTED
  O2: $p_2, d \leftarrow \text{TTest}(S_A, S_{\text{ref}}), \text{CohenD}(S_A, S_{\text{ref}})$
  if $p_2 > 10^{-6}$ **or** $|d| < 2.0$ **then**
    return REJECTED
  end if
  O3: $r \leftarrow \text{mean}(S_{\text{ref}})/\text{mean}(S_A)$; **if** $r < 5.0$ **return** REJECTED
  return VERIFIED

5 Experiments

5.1 Experimental Setup

We evaluate our framework on four datasets spanning different resolutions and scales: CIFAR-10 and CIFAR-100 (both 32×32, 50k training images with 5k reserved as the watermark set), STL-10 (96×96, 5k training images with 1k as watermark), and CelebA (64×64, 162k training images with 5k as watermark). All splits are deterministic and stratified with a fixed seed to ensure reproducibility.

For each dataset, we train a DDIM model using a U-Net backbone with 128 base channels, a cosine noise schedule, and the AdamW optimizer at a learning rate of 2×10^{-4} with EMA decay 0.9999. Training runs for 400k iterations in all cases. The Gaussian QR component uses an ensemble of B=50 models, each trained on a bootstrap sample of the auxiliary non-member data. Table 1 summarizes the key hyperparameters; full architectural and optimization details are deferred to Appendix A.

As public baselines for the verification protocol, we use pretrained checkpoints from HuggingFace: `google/ddpm-cifar10-32` serves as the reference for both CIFAR-10 and CIFAR-100, `google/ddpm-ema-bedroom-256` (resized) for STL-10, and `google/ddpm-celebahq-256` for CelebA. These baselines are trained on disjoint data from our watermark sets and represent the null hypothesis in the three-point verification protocol: a model that has never seen the owner's private evidence should exhibit substantially higher reconstruction error on that set.

Table 1: Key hyperparameters for reproducibility.

Component	Parameter	Value
DDIM	Iterations	400k
	Batch size / LR	128 / 2×10^{-4}
	EMA decay	0.9999
Gaussian QR	Ensemble B	50
	Bootstrap ratio	0.8
	Epochs / patience	50 / 10

Table 2: Ownership verification results. T-error (mean) on watermark set. All $p < 10^{-100}$ for baseline separation.

Dataset	Model	T-Error	$\lvert d\rvert$ / Ratio
CIFAR-10	Owner	28.6	23.9 / 24.4×
	Model B	28.6	23.9 / 24.4×
	Baseline	697.6	—
CIFAR-100	Owner	33.2	18.5 / 19.2×
	Model B	33.2	18.5 / 19.2×
	Baseline	636.6	—
STL-10	Owner	70.7	33.4 / 101×
	Model B	71.2	33.4 / 100×
	Baseline	7152.9	—
CelebA	Owner	63.5	26.1 / 26.6×
	Model B	63.5	26.1 / 26.6×
	Baseline	1691.8	—

Table 3: Membership inference performance using Gaussian QR margins. TPR computed at fixed FPR with 95% bootstrap CI (200 iterations).

Dataset	AUC	TPR@0.1%	TPR@0.01%
CIFAR-10	0.987 ± 0.002	0.89 ± 0.03	0.71 ± 0.05
CIFAR-100	0.982 ± 0.003	0.85 ± 0.04	0.65 ± 0.06
STL-10	0.968 ± 0.005	0.78 ± 0.05	0.55 ± 0.07
CelebA	0.975 ± 0.004	0.82 ± 0.04	0.60 ± 0.06

5.2 RQ1: Verification Effectiveness

Can private member evidence sets distinguish owner models from public baselines?

Table 2 reports t-error scores on the watermark set for owner models, fine-tuned derivatives (Model B), and public baselines.

Results: All datasets satisfy the three-point criteria. Separation: Cohen's d ranges from 18.5 (CIFAR-100) to 33.4 (STL-10), exceeding the $d > 2.0$ threshold by an order of magnitude. Consistency: Model $\mathcal{M}_A$ and B show statistically indistinguishable t-error ($p > 0.05$ for all datasets). Ratio: ranges from 19.2× to 101×, well above the 5× threshold.

Table 3 reports per-sample membership inference accuracy using Gaussian QR margins.

Table 4: Verification pass rate across threshold choices.

Threshold	C-10	C-100	STL	CelebA
d>1.5, r>5×	✓	✓	✓	✓
d>2.0, r>5× (ours)	✓	✓	✓	✓
d>3.0, r>10×	✓	✓	✓	✓

Threshold Sensitivity: The thresholds $d > 2.0$ and ratio $> 5\times$ are conservative. Table 4 shows that all datasets pass verification even under stricter thresholds (up to $d > 3.0$, ratio $> 10\times$), confirming that results are not artifacts of threshold tuning.

5.3 RQ2: Post-Theft Robustness

A practical adversary who obtains a stolen model will typically fine-tune it to a new domain or task before deployment. The ownership signal must survive such adaptation. To evaluate this, we simulate post-theft modification via MMD fine-tuning [**?**], which adapts Model A to minimize distribution mismatch in CLIP feature space while preserving generation quality. Starting from Model A, we fine-tune for 500 iterations using 10-step DDIM sampling, a cubic polynomial kernel computed over CLIP ViT-B/32 features, a learning rate of 5×10^{-6}, and EMA decay 0.9999.

The resulting Model B retains nearly identical t-error to Model A on the watermark set (Table 2), and all three verification criteria remain satisfied across every dataset. This robustness is not coincidental: MMD fine-tuning optimizes the distribution of *newly generated* samples to match a target domain, but does not explicitly minimize reconstruction error on existing training data. The memorization signal encoded in the model's response to watermark samples is therefore largely orthogonal to the fine-tuning objective, explaining why it persists even after substantial adaptation. We note, however, that an adversary who retrains from scratch or specifically targets memorization erasure falls outside our threat model .

5.4 Controlled Baseline Comparison

Setup. Tables 5 and 6 compare MiO against two representative diffusion-model watermarking baselines under controlled conditions. All methods are trained on CIFAR-10 using identical data (50K images) and evaluated under the same robustness protocol using their official codebases. WDM [**?**] embeds a watermark diffusion process into a DDPM backbone; Zhao et al. [**?**] watermark training images via StegaStamp before training an EDM backbone. DeepMarks [**?**] is excluded because its weight-PDF fingerprinting targets discriminative classifiers and does not transfer to diffusion U-Nets in a principled way: modern diffusion U-Nets interleave residual convolutions, self-attention, and group normalization—layer types whose weight distributions resist uniform GMM shaping. Generation quality is measured by FID (50K samples, each method's native sampler). ΔFID reports the quality

Table 5: Controlled comparison with watermarking baselines on CIFAR-10. All methods trained on identical data (50K images). †Uses EDM backbone (not DDPM/DDIM). ΔFID: difference from unwatermarked baseline. Train OH: additional training cost relative to clean model.

Method	Native Metric	FID↓	ΔFID	Train OH
Clean (no WM)	—	–	0	—
WDM	WM ext. –%	–	–	∼1×
Zhao†	Bit acc. –%	–	–	∼1×
MiO (Ours)	TPR 89%@0.1%	–	**0.0**	**0**

Table 6: Robustness on CIFAR-10. Each cell: native verification pass (✓) or fail (✗), with FID in parentheses. MMD-FT: 500 iterations, lr = 5e-6, CLIP features. Prune: 30% structured L1 pruning. †EDM backbone.

Method	Clean	MMD-FT	Prune 30%
WDM	– (–)	– (–)	– (–)
Zhao†	– (–)	– (–)	– (–)
MiO	– (–)	– (–)	– (–)

gap relative to an unwatermarked model trained on the same data.

Key findings. (i) Both baselines achieve near-perfect native verification on clean models, consistent with their published results. (ii) MiO is the only method with *zero training overhead* and no FID degradation, since it operates post-hoc on an unmodified model. (iii) MiO's per-sample TPR at FPR = 0.1% (89%) is lower than watermarking decode rates (>99%), but these quantities are not comparable: watermark decode measures recoverability of an injected signal, while MiO's TPR is calibrated under an explicit false-positive budget without any model modification. Table 6 evaluates robustness under MMD fine-tuning (500 iterations, lr = 5e-6, CLIP features) and 30% structured L1 pruning. We additionally evaluate whether an adversary can retroactively fabricate a MiO ownership claim; see Appendix F for the full analysis.

5.5 RQ3: Score Stability and Threshold Calibration

The verification framework involves several design choices—how to aggregate multi-timestep evidence, how many timesteps to sample, and how to model the score distribution for threshold calibration. We ablate the most consequential choices here; additional analyses of ensemble size and computational trade-offs are deferred to Appendix B.

Aggregation strategy. Table 7 compares four aggregation strategies applied to the t-error sequence on CIFAR-10. The 25th percentile (Q25) achieves the strongest separation, with Cohen's $d = -23.93$ and a ratio of 24.4×, outperforming both the mean ($d = -18.2$) and the median ($d = -19.8$). The pattern is intuitive: lower percentiles isolate the timesteps at which the memorization gap is most pronounced—noise levels where the model reconstructs its own training data far more accurately than unseen samples. However, going too low (Q10) introduces sensitivity to outlier timesteps, yielding slightly weaker separation than Q25. The 25th percentile strikes an effective balance between capturing the strongest memorization signal and suppressing per-timestep noise.

Table 7: Ablation: Aggregation method on CIFAR-10. Q25 achieves the strongest separation from public baselines.

Aggregation	Cohen's d	Ratio
Mean	−18.2	15.1
Q10	−21.5	20.8
Q25 (ours)	**−23.93**	**24.40**
Median	−19.8	17.2

Number of timesteps. We vary the number of sampled timesteps K from 10 to 100 on CIFAR-10 and find that separation ($|d|$) stabilizes once $K \geq 25$, with negligible improvement beyond $K = 50$. This suggests that 50 uniformly sampled timesteps are sufficient to capture the memorization signal reliably, and we adopt K=50 as the default across all experiments. Practitioners operating under tighter computational budgets can reduce K to 25 with minimal loss in verification power.

Gaussian vs. pinball-loss quantile regression. A natural alternative to our Gaussian QR formulation is pinball-loss quantile regression, which directly optimizes for a specific quantile τ without distributional assumptions. We compare the two approaches on CIFAR-10 at $\tau \in \{0.999, 0.9999\}$ (corresponding to FPR = 0.1% and 0.01%). Gaussian QR achieves comparable TPR at both operating points while offering two practical advantages: first, a single trained model supports arbitrary τ via the closed-form expression in Eq., eliminating the need to retrain for each desired FPR level; second, the total parameter count is reduced by roughly half, since pinball QR requires a separate model for every quantile of interest. These properties make Gaussian QR substantially more convenient for deployment, where the verifier may wish to evaluate ownership claims at multiple significance levels without retraining the scoring model.

5.6 Computational Cost

Table 8 compares the computational overhead of MiO against pinball-loss quantile regression and representative watermarking baselines. MiO's primary offline cost is training the Gaussian QR ensemble, which totals approximately 100 A100-hours for B=50 but parallelizes trivially (wall-clock ∼2 h with 50 GPUs). At inference, a complete ownership query—from t-error scoring on 5,000 watermark samples through ensemble prediction to the three-point verdict—completes in under 15

Table 8: Computational cost comparison on CIFAR-10 (NVIDIA A100).

Method	Training Cost	Inference	Parameters
Ownership verification (ours)			
Gaussian QR (B=50)	∼100 h†	<1 min	11.2M × 1
t-error scoring	—	∼10 min	—
Full query	—	∼11 min	—
Alternative QR formulation			
Pinball QR ($\lvert\mathcal{T}\rvert$=5)	∼10 h × 5	<1 min	11.2M × 5
Explicit watermarking			
WDM*	Full retrain	Decode	Modified model
Zhao et al.*	Full retrain	Bit extract	Modified model

† Embarrassingly parallel; wall-clock ∼2 h with 50 GPUs.

* Requires retraining the diffusion model with watermark objective.

minutes on a single A100. Compared to pinball-loss QR, the Gaussian formulation reduces total parameters by roughly $|\mathcal{T}|\times$ (one model serves all quantile levels), and compared to explicit watermarking methods, MiO imposes no cost on the owner model's training pipeline itself.

6 Discussion and Limitations

We frame scope conditions rather than limitations.

White-box Access: Our method applies where model weights are available, standard in IP litigation (courts compel disclosure), regulatory audits (GDPR/AI Act), partnership disputes, and increasingly common open-weight models (LLaMA, Mistral).

Watermark Retention: Verification requires the owner to retain the watermark set, analogous to keeping receipts for warranty claims.

Baseline Availability: Results are strongest with domain-matched public baselines; for novel domains, general-purpose baselines provide conservative bounds.

Gaussian Assumption: The Gaussian model provides a principled approximation; empirical calibration shows quantile estimates remain accurate under moderate misspecification.

What This Work Does NOT Address: API-only verification (we focus on white-box), full retraining adversaries (our threat model assumes fine-tuning), and adversaries specifically optimizing to remove memorization signals.

7 Conclusion

We presented a novel ownership verification framework for diffusion models using t-error based reconstruction disparity and Gaussian Quantile Regression. By formalizing the distinction between membership inference and ownership verification, we established that public baselines and statistical hypothesis testing are *necessary* components for ownership claims. Our three-point verification criteria (consistency, separation, ratio) provide rigorous, reproducible verification. Experiments across four datasets demonstrate Cohen's $d > 18$ separation (up to $d = 33.4$ on STL-10), with criteria satisfied even after adversarial MMD fine-tuning.

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A Additional Experimental Details

A.1 Hyperparameters

Table 9: Training hyperparameters for DDIM models.

Parameter	Value
Total iterations	400,000
Batch size	128
Learning rate	2×10^{-4}
Optimizer	AdamW
Weight decay	0.0
EMA decay	0.9999
Noise schedule	Cosine
Timesteps T	1000
U-Net Architecture	
Base channels	128
Channel multipliers	[1, 2, 2, 2]
Attention resolutions	[16]
Dropout	0.1

A.2 MMD Fine-tuning Configuration

Table 10: Gaussian QR training hyperparameters.

Parameter	Value
Backbone	ResNet-18 (pretrained)
Ensemble size B	50
Bootstrap ratio	0.8
Epochs per model	50
Batch size	256
Learning rate	0.001
LR schedule	Cosine annealing
Early stopping patience	10 epochs

Table 11: MMD fine-tuning hyperparameters.

Parameter	Value
Iterations	500
Learning rate	5×10^{-6}
Batch size	128
DDIM steps	10
t_{start}	900
EMA decay	0.9999
Kernel	Cubic polynomial
Feature extractor	CLIP ViT-B/32 (frozen)

B Additional Ablations

B.1 Ensemble Size

We vary the ensemble size B from 1 to 100 and measure the standard deviation of the predicted quantile $\hat{q}_\tau(x)$ across five independent runs on the CIFAR-10 watermark set at $\tau = 0.999$ (FPR $= 0.1\%$). Table 12 reports the results. Prediction variance drops sharply from $B = 1$ to $B = 20$, continues to decrease up to $B = 50$, and plateaus beyond that point. We adopt $B = 50$ as the default, which offers a favorable trade-off between stability and computational cost. Since ensemble members are trained independently, scaling from $B = 50$ to any larger value is embarrassingly parallel and requires no inter-model communication.

Table 12: Effect of ensemble size B on quantile prediction stability (CIFAR-10, $\tau = 0.999$). Std. dev. computed over 5 independent runs.

Ensemble B	Std($\hat{q}_\tau$)	TPR @ FPR 0.1%
1	0.142	0.83 ± 0.05
10	0.068	0.87 ± 0.03
20	0.041	0.88 ± 0.03
50 (ours)	0.019	0.89 ± 0.03
100	0.016	0.89 ± 0.02

B.2 Computational Cost

Table 13 summarizes the computational requirements of each component. The dominant cost is ensemble training (approximately 100 A100-hours for $B = 50$), which parallelizes trivially. At inference, t-error computation for 5,000 watermark samples takes roughly 10 minutes on a single A100, and ensemble quantile prediction adds less than 1 minute. In total, a single ownership query (from raw images to verdict)

Table 15: Verification pass rate across threshold choices. All datasets pass even with stricter thresholds.

Threshold	CIFAR-10	CIFAR-100	STL-10	CelebA
$d > 1.0$, ratio $> 3\times$	✓	✓	✓	✓
$d > 1.5$, ratio $> 5\times$	✓	✓	✓	✓
$d > 2.0$, ratio $> 5\times$ (ours)	✓	✓	✓	✓
$d > 2.5$, ratio $> 7\times$	✓	✓	✓	✓
$d > 3.0$, ratio $> 10\times$	✓	✓	✓	✓

Algorithm 3 Gaussian Quantile Regression Training

Require: Auxiliary non-member data $\mathcal{D}_{\text{aux}}$, ensemble size B, bootstrap ratio r
Ensure: Trained ensemble $\{f_\theta^{(b)}\}_{b=1}^{B}$
  for $b = 1$ to B **do**
    $\mathcal{D}_b \leftarrow \text{BootstrapSample}(\mathcal{D}_{\text{aux}}, r)$ {80% with replacement}
    $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}} \leftarrow \text{Split}(\mathcal{D}_b, 0.9)$
    Initialize $f_\theta^{(b)}$ with ResNet-18 backbone
    repeat
      for batch $(x, s(x), \phi(x))$ in $\mathcal{D}_{\text{train}}$ **do**
        $y \leftarrow \log(1 + s(x))$
        $(\mu, \log\sigma) \leftarrow f_\theta^{(b)}(x, \phi(x))$
        $\mathcal{L} \leftarrow (y - \mu)^2/(2\sigma^2) + \log\sigma$
        Update θ via gradient descent
      end for
    until early stopping (patience = 10)
  end for
  return $\{f_\theta^{(b)}\}_{b=1}^{B}$

Table 16: Retroactive claim defense on CIFAR-10. An adversary attempts to construct a fake watermark set $\mathcal{W}'$ that passes the three-point verification protocol. Cohen's d and ratio are computed against public baselines. Three-point: all three verification criteria (Sec. 5.4) satisfied (✓) or not (✗).

Adversary Strategy	Cohen's d	Ratio	Three-point
True owner ($\mathcal{W}$, 5K)	–	–	–
Random 5K (mean of 100 draws)	–	–	–
Cherry-picked top-5K	–	–	–
Top-5K lowest t-error	–	–	–
Non-member (test set, 5K)	–	–	–
Wrong model	–	–	–

completes in under 15 minutes, making the protocol practical for on-demand verification.

By comparison, explicit watermarking methods such as WDM [?] require re-training the diffusion model with the watermark objective, which typically takes as long as the original model training (on the order of days for the architectures we consider). MiO's Gaussian QR training is a one-time offline cost, and the owner model itself is never modified.

Table 13: Computational cost breakdown on CIFAR-10 (single NVIDIA A100).

Component	Time	Parameters
DDIM training (owner model)	∼48 h	35.7M
Gaussian QR (×1 model)	∼2 h	11.2M
Gaussian QR ensemble (B=50)	∼100 h†	560M total
Inference (per verification query)		
t-error scoring (5k samples)	∼10 min	—
Ensemble prediction	<1 min	—
Three-point criteria	<1 s	—
Total query time	**∼11 min**	—

† Embarrassingly parallel; wall-clock ∼2 h with 50 GPUs.

C Perceptual Hash Evaluation

Table 14: Perceptual hash (pHash) performance on CIFAR-10 train vs. test set deduplication. FPR: false positive rate (distinct images flagged as duplicates). FNR: false negative rate (actual duplicates missed).

Hamming Threshold	FPR (%)	FNR (%)	Flagged Pairs
3 bits	0.02	5.1	12
5 bits (ours)	0.10	2.0	48
7 bits	0.35	0.8	175

We use threshold of 5 bits as a balance between FPR and FNR. Images are upsampled to 64×64 before hashing due to CIFAR's low native resolution.

D Threshold Sensitivity Analysis

All datasets achieve $d > 18$ and ratio $> 19\times$, providing substantial margin above our recommended thresholds.

E Gaussian QR Training Details

F Retroactive Claim Defense

An adversary who observes a published model could attempt to retroactively fabricate an ownership claim by selecting a subset of training data that happens to exhibit low t-error. Table **??** evaluates this attack under several adversary strategies on CIFAR-10.